A Two-System Framework for Truth-Grade Three-Body Simulation and Sparse Discovery of Interpretable Predictive Dynamics

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Abstract

This project is guided by a central question posed at project inception by Jonathan Hendler: can richer sensing/information in a deterministic three-body system enable mathematically better prediction over time? We present a two-system framework for testing that question under coupled gravitational and electromagnetic interactions. The first system is a high-fidelity numerical simulator ("oracle") designed for reproducibility, performance, and explicit error control. It supports Newtonian gravity and a nonrelativistic electromagnetic interaction model including electric and magnetic fields as well as scalar and vector potentials (ϕ, A). The second system is a model-discovery engine that searches for sparse, interpretable equations that predict the oracle's trajectories with minimal complexity. In the current implementation, discovery proceeds via structured feature libraries and sparse regression (STLS/LASSO)[1, 5], using simulator-provided accelerations to avoid derivative noise; an optional GA baseline is also available.

This revision adds three math families outside the prior force-library core (`jacobi_legendre`, `shape_invariants`, `symmetry_invariants`), plus an MDL-aware deterministic selector mode (`numeric_mdl`) layered on top of existing MCTS archive reuse and strict holdout gates. In matched deterministic gravity-only runs on February 8, 2026 (`research_v2_atlas`, 3 iterations, 150 steps, $dt = 0.01$), the best equation and benchmark metrics were unchanged versus baseline: median relative improvement 0.0 (12-seed suite, 95% bootstrap CI $[0, 0]$), benchmark relative improvement $\approx 1.3 \times 10^{-16}$, and `strict_holdout_passed=false`. A targeted global-model run with the new families produced a simpler equation (complexity 5) but a small negative benchmark delta (-7.5×10^{-5}, CI fully negative). Therefore, current evidence supports tooling progress and stronger falsification, not a new physical law.

We additionally implemented a deterministic delay-embedding track (`predictability takens`) with strict chronological holdouts, local deterministic formulas, and a sensitivity-consistency score based on one-step tangent prediction. Beyond local linear and local rational families, we added residual (delta) variants that fit $\Delta x_{t+1} = x_{t+1} - x_t$ and reconstruct $x_{t+1} = x_t + \widehat{\Delta x}_{t+1}$. On a matched 4-channel benchmark (February 8, 2026), the delta-enabled baseline (target-only embedding) was mixed: 2/4 channels effective, median all-channel improvement ≈ 0.23797 (95% bootstrap CI $[-0.52325, 0.99981]$). With sensor-augmented embeddings plus delta-enabled model selection, efficacy rose to 4/4 effective channels with median all-channel improvement ≈ 0.99989 (95% CI $[0.99980, 0.99999]$), and paired before/after comparison showed strict non-regression on all shared channels. To make the guiding question operational, we added a reporting layer that now includes deterministic bootstrap confidence intervals and an explicit before/after comparator with effect-size and non-regression flags. The measured gain from baseline to sensor-augmented delta mode was substantial on this run: all-channel median improvement delta $\approx +0.7619$ and derived-channel median improvement delta $\approx +1.5231$. In a same-day follow-up stress test on a separate perturbed-three-body trajectory, sensor gains

were non-monotonic: one `min_pair_dist` target-only scan reached a strict high-bar pass at $W^\star = 4$, while richer position/state sensor sets did not pass in the tested grid. We also introduced an information-context framing: for a target channel and sensor set, define context volume $W = (\tau m)|S|$ and estimate the minimum $W^\star$ that meets a strict holdout gate. In a matched high-bar analysis ($\Delta_{\mathrm{rel}} > 0.6$, sensitivity median ≤ 0.2), we measured $W^\star = 3$ for one-body inertial and two-body Kepler regimes, versus $W^\star = 20$ for a perturbed three-body regime, suggesting higher information demand as interaction complexity increases.

We emphasize editorially the areas of modeling uncertainty: the electromagnetic approximation (quasi-static, no retardation or radiation reaction)[20], conservation-law interpretation under the chosen EM model, close-encounter numerical stiffness, and the limits of global predictability in chaotic regimes[6]. Relative to bounded-horizon ANN surrogates for three-body dynamics[7] and generic sparse-discovery pipelines[1], novelty here is methodological: deterministic claim-gated reporting, explicit information-context thresholds, and reproducible failure reporting against the oracle. In matched February 10, 2026 head-to-head runs on `a1_x`, the new `delta_tcn` neural residual outperformed `delta_mlp` in both single-sensor and sensor-augmented settings, but current breadth remains insufficient for a physics-grade novelty claim. Code repository: `https://github.com/Hendler/threebody`.

1 Introduction

The classical three-body problem is a paradigmatic nonlinear dynamical system exhibiting sensitivity to initial conditions and, for most configurations, the absence of closed-form global solutions. Practical prediction therefore relies on numerical integration, while theory offers partial analytic structure (integrals of motion, special solutions, perturbation expansions, invariant manifolds). For modern applications—ranging from celestial mechanics to charged particle dynamics—one often wants both: (i) an accurate simulator suitable as "ground truth," and (ii) interpretable approximations that predict well and illuminate structure.

Guiding question and scope

At project inception, Jonathan Hendler posed the guiding question:

> Given richer measurements of a deterministic three-body system, can we discover and use mathematically better predictive laws?

All methods and evaluations in this paper are judged against that question: they must improve strict holdout prediction, not only in-sample fit.

Thinking behind this work

The design of this project follows four decisions.

- **Deterministic does not mean easy:** chaotic three-body dynamics are not random, but sensitivity makes long-horizon microstate prediction fragile.
- **Information should be testable as value:** if better sensors help, they must improve strict holdout metrics, not only fit on seen data.
- **Model class must match channel structure:** richer inputs can help only when the equation family can use them without overfitting.
- **Negative outcomes are scientific outcomes:** failures (no improvement, or degradation) are explicitly reported and claim-gated.

This is why we use deterministic protocols (fixed splits/seeds), paired before/after comparisons, and conservative claim gates.

Relation to prior work and novelty boundary

Related literature gives useful anchors: data-driven sparse equation discovery (e.g., SINDy)[1]; delay-embedding reconstruction and embedding-dimension diagnostics[8, 9, 10]; and bounded-horizon neural surrogates for chaotic three-body trajectories[7]. Our contribution is not a universal closed-form solution and not a claim that local models solve long-horizon chaos. The genuinely new element is the integrated protocol: strict claim gates tied to holdout metrics, explicit context-volume testing via $W = (\tau m)|S|$, and deterministic before/after sensor-value comparisons.

Predictability context-window framing

To sharpen the guiding question, we treat prediction as an information-budget problem. For a target channel x_t and sensor set S, a delay embedding uses:

$$z_t = \{x^{(s)}_{t-j\tau} : j = 0, \ldots, m-1,\ s \in S\},$$

with context volume

$$W = (\tau m)\,|S|.$$

Given a strict effectiveness gate (τ_Δ, τ_s) on holdout relative improvement and sensitivity consistency, we define the *minimum effective context*:

$$W^\star = \min\{W : \Delta_{\mathrm{rel}}(W) > \tau_\Delta,\ \varepsilon_{\mathrm{sens}}(W) \le \tau_s\}.$$

If no tested context passes, we report "not found in tested grid" rather than claiming failure of determinism. This framing connects directly to Jonathan Hendler's primary question: better sensors and better mathematical representations should reduce forecast error by moving the model into an effective context regime.

We propose a two-system approach:

- **Truth-grade simulation system ("oracle")** – a careful, reproducible, numerically controlled engine that produces trajectories and physically meaningful diagnostics.
- **Equation discovery system ("predictor")** – a sparse identification pipeline that produces compact, interpretable predictive models that approximate the oracle and are validated by forward rollout.

This paper describes the architecture, numerical methods, and discovery methodology, and it explicitly lists modeling assumptions and unresolved issues.

Reader guide (physicists and novice readers)

For physicists: the main body focuses on the central question, methods, and claim-level outcomes; implementation details and extended ablations are moved to appendices for auditability. For novice readers: the workflow is "simulate first, then discover" — we generate trusted trajectories with a strict simulator, ask sparse regression/search to propose simple equations, and only keep equations that roll out accurately on new tests.

2 Simulator ("Oracle") Design

2.1 State, coordinate choices, and invariants

The simulator evolves three bodies indexed by $i \in \{1, 2, 3\}$ with:

- Mass m_i, charge q_i.
- Position $r_i(t) \in \mathbb{R}^3$.
- Velocity $v_i(t) = \dot{r}_i(t) \in \mathbb{R}^3$.

The current Rust implementation is fixed to three bodies ($N = 3$) across simulation, output schema, and discovery; general N-body support is future work.

We typically re-center initial conditions to the center-of-mass (COM) frame and remove net linear momentum (barycentric initialization) to reduce drift. Invariants used as diagnostics include:

- Total linear momentum $P = \sum_i m_i v_i$.
- Total angular momentum $L = \sum_i r_i \times (m_i v_i)$.

A "mechanical energy" proxy (Section 2.4) is logged, with editorial caveats for EM.

2.2 Interaction model: gravity and nonrelativistic electromagnetism

Gravity (Newtonian). We use

$$a_i^{(g)} = \sum_{j \neq i} G m_j \frac{r_j - r_i}{\|r_j - r_i\|^3}.$$

Electromagnetism (chosen approximation). We include an explicit, nonrelativistic EM model.

Electric field at particle i:

$$E_i = \sum_{j \neq i} k_e q_j \frac{r_i - r_j}{\|r_i - r_j\|^3}.$$

Magnetic field at particle i via a quasi-static Biot–Savart-like term for moving charges:

$$B_i = \sum_{j \neq i} \frac{\mu_0}{4\pi} q_j \frac{v_j \times (r_i - r_j)}{\|r_i - r_j\|^3}.$$

Lorentz acceleration:

$$a_i^{(em)} = \frac{q_i}{m_i} \left(E_i + v_i \times B_i \right).$$

Total acceleration:

$$a_i = a_i^{(g)} + a_i^{(em)}.$$

We also compute potentials.

Scalar potential:

$$\phi_i = \sum_{j \neq i} k_e \frac{q_j}{\|r_i - r_j\|}.$$

Vector potential:

$$A_i = \sum_{j \neq i} \frac{\mu_0}{4\pi} \frac{q_j v_j}{\|r_i - r_j\|}.$$

Editorial uncertainty: This EM model omits finite propagation speed (retardation), radiation reaction, and self-field effects. It is best interpreted as a quasi-static approximation[20]; its domain of physical validity is not universal. For "truth-grade" claims, the simulator is truth-grade relative to the assumed equations, not necessarily relative to full Maxwell–Lorentz dynamics.

2.3 Numerical integration

We provide two integrators:

- **Adaptive RK45 (Dormand–Prince)**[12]: used as "truth mode" for controlled local truncation error. It adapts time step using embedded error estimates with user-specified tolerances and min/max dt.
- **Leapfrog (Kick–Drift–Kick)**[13, 14, 16, 15]: a fast, stable second-order method often favored for long-horizon gravitational dynamics.

We implement a strict "truth mode" that enables adaptive RK45 with step rejection and per-step dt tracking. If the embedded error exceeds tolerance, the step is rejected and dt is reduced; otherwise, dt is increased within configured bounds. This provides a best-effort accuracy strategy under the stated model assumptions, especially near close encounters where fixed-step methods can fail.

Editorial uncertainty: For purely Hamiltonian gravity, symplectic methods have strong long-term stability guarantees. With the chosen EM terms, the system is not treated with a full Hamiltonian field formulation here; leapfrog remains useful but should not be assumed to preserve the same structure exactly. This is one reason we maintain RK-based truth mode.

2.4 Diagnostics and conservation interpretation

We log $P(t)$ and $L(t)$, plus a "mechanical energy" diagnostic:

- Kinetic: $\sum_i \frac{1}{2} m_i \|v_i\|^2$.
- Gravitational potential: $\sum_{i<j} -G m_i m_j / \|r_{ij}\|$.
- Electric potential: $\sum_{i<j} k_e q_i q_j / \|r_{ij}\|$.

where $r_{ij} = r_j - r_i$.

Editorial uncertainty: In full electrodynamics, energy conservation includes field energy and radiation; in quasi-static approximations, interpreting "total energy" is subtle. We therefore treat energy drift as a numerical health metric, not a strict physical invariant under all EM configurations.

2.5 Close encounters and stiffness

Near-collisions can cause:

- Accelerations scaling as $1/r^2$ (and worse in some derived terms).
- Adaptive dt collapse.
- Large floating-point conditioning issues.

Current implementation supports optional softening ϵ in denominators to avoid singularities. In addition, when a close-encounter policy triggers softening, we can (optionally) ramp ϵ smoothly as a function of the minimum pair distance (to avoid discontinuous force jumps), and we can deterministically substep fixed-step integrators near close approaches (to reduce stiffness without changing the global output cadence). Encounter events record the minimum pair distance, the action taken, and (when applicable) the softening change and substeps used as part of the sidecar metadata.

Editorial uncertainty: These are pragmatic numerical stabilizers and still change the effective physics. A truly "truth-grade" approach for close encounters would incorporate regularization (e.g., KS/chain regularization)[21, 22] and/or higher precision. This remains a clear future-work item.

2.6 Output contract for downstream discovery

The simulator outputs a wide CSV including:

- r_i, v_i for each body.
- a_i (total acceleration) for each body.
- E_i, B_i, ϕ_i, A_i.
- Diagnostics: energy proxy, P, L, drifts.

Design rationale: Discovery should use simulator-provided accelerations, not finite differences of noisy trajectories.

3 Discovery Engine ("Predictor"): Sparse, Interpretable Dynamics

3.1 Objective

We seek simple, interpretable functions $\hat{f}_i$ that approximate oracle accelerations, building on work in automated equation discovery and sparse model identification[3, 4, 1]:

$$a_i(t) \approx \hat{f}_i\left(\text{features}(r(t), v(t), \ldots)\right),$$

with high forward-prediction fidelity over a time horizon, minimal equation complexity, and explicit domain-of-validity.

This is fundamentally different from "solving the three-body problem in closed form." The goal is regime-specific compact predictive structure. Related work extends sparse discovery to spatiotemporal systems and PDE settings.[2]

3.2 Vector targets and component-wise discovery

In implementation, we fit acceleration components separately (e.g., $a_{i,x}$, $a_{i,y}$, $a_{i,z}$) using a shared feature library, then combine the component equations into a vector-valued model. This preserves interpretability while enabling full trajectory rollout evaluation.

3.3 Why we fit acceleration directly

Classical system identification often estimates $\dot{x}$ from measurements; in chaotic systems this introduces strong noise amplification. We avoid this by training on simulator-provided $a_i(t)$ rather than differentiating $v_i(t)$.

3.4 Feature libraries (Θ) as structured hypothesis spaces

We build vector-valued basis functions ("terms") from physically meaningful atoms:

- Relative vectors $r_{ij} = r_j - r_i$.
- Distances $d_{ij} = \|r_{ij}\|$.
- Unit directions $\hat{r}_{ij}$.
- Velocities and differences v_i, v_{ij}.
- Cross products capturing magnetic-like structure $v_j \times r_{ij}$.
- Triple products capturing Lorentz-like structure $v_i \times (v_j \times r_{ij})$.

We implement libraries of increasing expressivity:

- **default** (current implementation): direction-preserving, component-wise bases that mirror the analytic force contributions *without* physical constants:
  - Gravity basis: $g_i^* = \sum_{j \neq i} m_j (r_j - r_i)/\|r_j - r_i\|^3$.
  - Electric basis: $e_i^* = (q_i/m_i) \sum_{j \neq i} q_j (r_i - r_j)/\|r_i - r_j\|^3$.
  - Magnetic basis: $b_i^* = (q_i/m_i)\, v_i \times B_i^*$, where $B_i^* = (1/4\pi) \sum_{j \neq i} q_j (v_j \times (r_i - r_j))/\|r_i - r_j\|^3$.

  Under our quasi-static model, this yields $a_i \approx G\, g_i^* + k_e\, e_i^* + \mu_0\, b_i^*$ (with softening applied consistently).
- **extended** (current implementation): adds a small number of carefully chosen atoms to expand expressivity while preserving interpretability, including (i) additional distance scalings (e.g., $1/r^4$ variants), and (ii) simple distance-based gating ("close" vs. "far" features) to help isolate regime-specific structure without resorting to an unconstrained polynomial soup.
- **em_fields** (current implementation): a minimal EM library that learns Lorentz acceleration directly from simulator fields. Features are in acceleration units (already include physical constants), so in EM-enabled runs we expect coefficients near 1:
  - Electric term: $\ell_{E,i} = (q_i/m_i) E_i$.
  - Magnetic term: $\ell_{v \times B,i} = (q_i/m_i)(v_i \times B_i)$.

  This supports a simple discovered form $a_i \approx G\, g_i^* + 1 \cdot \ell_{E,i} + 1 \cdot \ell_{v \times B,i}$ (with consistent softening).

Editorial note: The library design is where most "new math" leverage lives. If the library is too broad, discovery becomes unstable and non-interpretable; too narrow, it cannot capture dynamics. We currently err toward small, interpretable libraries.

3.5 Sparse discovery via sparse regression (current implementation)

Let y be a target acceleration component (e.g., $a_{i,x}$) and let Θ be the feature matrix evaluated over time (columns are library terms evaluated at each time step). We restrict to models linear in features,

$$y \approx \Theta \xi,$$

and in the current Rust implementation we estimate ξ on a fixed library via sparse regression (retaining the top-3 equations per component across a small hyperparameter path):

- **STLS** (sequential thresholded least squares): normalize library columns, solve ridge/OLS for ξ, hard-threshold small coefficients below τ, refit on the surviving support, and iterate until the support stabilizes.
- **LASSO**: coordinate descent along an ℓ_1 path parameterized by α (warm-started), yielding sparse ξ on a fixed library.
- A lightweight genetic algorithm (GA) over sparse coefficient vectors remains available as an optional baseline; it is slower and non-deterministic but provides an alternative search dynamic.

In addition to per-component coefficient estimation, the CLI's `factory` loop maintains a small *population* of full vector equations across iterations: it carries forward the incumbent best and a few recent elites, injects a handful of axis-consistent mutations (an equation-level GA), and re-scores them under new initial conditions. This makes candidate equations the primary object of iteration while keeping numeric rollout metrics as the selection criterion.

Related work: Sparse regression (SINDy/STLS)[1] and LASSO-style solvers[5] are natural coefficient estimators for fixed libraries and are implemented in the Rust codebase; the data contract (header-based CSV plus sidecar metadata) supports reproducible regression and search-based baselines.

3.6 Archive-guided equation search across iterations (current implementation)

To retain and reuse successful equation structures across iterations, we added a persistent equation archive and an MCTS-style parent selector in the `factory` loop.

For each candidate equation e, we define a scalar score

$$s(e) = \mathrm{RMSE}_{\mathrm{rollout}}(e) + 0.1 \log(1 + \mathrm{MSE}(e)) + 10^{-3}\,\mathrm{complexity}(e).$$

Each iteration updates an archive with the top-K candidates (default $K = 12$), storing per-equation visit count $n(e)$, mean score $\bar{s}(e)$, best score $s^\star(e)$, score dispersion estimate $\hat{\sigma}(e)$, and seed-coverage counts.

Archive parent selection uses a UCT-like priority

$$u(e) = \frac{1}{1 + s^\star(e)} + c\sqrt{\frac{\log(1+N)}{1+n(e)}} + \beta\frac{\hat{\sigma}(e)}{1+\bar{s}(e)},$$

where N is the archive total visit count, c is an exploration constant (default $c = 0.4$), and β is an uncertainty bonus (default $\beta = 0.2$ under the `research_v1` policy). The top-M archive parents (default $M = 2$) are injected into the equation-mutation pool each iteration after normalization and deduplication.

We also added an active initial-condition heuristic: when multiple top equations exist, we score candidate IC seeds by disagreement among their predicted $t = 0$ accelerations and preferentially sample higher-disagreement seeds. This improves equation discrimination pressure without changing the final numeric ranking rule.

This adds explicit search memory and reproducible artifacts (`equation_search_archive.json`, `equation_search_report.md`, `policy_effective.json`) without changing the primary selection criterion (numeric rollout metrics).

3.7 Local deterministic atlas + sensitivity equation (new in this revision)

To target regime-local structure, we added a model-family switch:

- **global**: one sparse equation per acceleration component.
- **atlas**: separate sparse equations on close/far gated subsets, then merged into a deterministic piecewise formula.

In the current implementation, atlas gating supports two deterministic modes selected by `-atlas-gate`:

- **binary**:
$$\mathrm{gate}_{\mathrm{close}} = \mathbf{1}\{r_{\min} < r_{\mathrm{gate}}\}, \qquad \mathrm{gate}_{\mathrm{far}} = 1 - \mathrm{gate}_{\mathrm{close}},$$
where $r_{\min} = \min_{i<j} \|r_{ij}\|$.
- **smooth** (current default):
$$\mathrm{gate}^{(s)}_{\mathrm{close}} = \sigma\left(\frac{r_0 - r_{\min}}{w}\right), \qquad \mathrm{gate}^{(s)}_{\mathrm{far}} = 1 - \mathrm{gate}^{(s)}_{\mathrm{close}},$$
with user-set center r_0 and width $w > 0$.

For smooth mode we additionally include two shifted smooth pairs

$$\mathrm{gate}^{(s,\Delta)}_{\mathrm{close}} = \sigma\left(\frac{(r_0 + \Delta) - r_{\min}}{w}\right), \qquad \mathrm{gate}^{(s,\Delta)}_{\mathrm{far}} = 1 - \mathrm{gate}^{(s,\Delta)}_{\mathrm{close}},$$

with $\Delta \in \{-w, +w\}$, yielding three deterministic smooth atlas variants (`primary`, `inner`, `outer`). The library includes corresponding gated atoms (e.g., $g^{\mathrm{close}}_{i,\alpha}$, $g^{\mathrm{far}}_{i,\alpha}$ and smooth/multi-smooth variants), and can be augmented with explicit hypothesis families (`pn1`, `yukawa`, `tidal_invariants`, `jerk_augmented`, `hamiltonian_invariants`).

For smooth atlas runs, we also implement a deterministic one-shot gate-geometry tuner on the first trajectory. It chooses (r_0, w) from a small grid by maximizing

$$J(r_0, w) = 0.65\, B + 0.35\, C,$$

where B is a gate-balance score (close/far occupancy near 50%) and C is a close-vs-far acceleration-contrast score; the selected values and candidate scores are logged in `gate_geometry.json`.

To test whether discovered dynamics encode meaningful local linearization, we added a tangent-flow diagnostic on the discovered predictor:

$$\dot{x} = f_\theta(x), \qquad \dot{\delta x} = J_{f_\theta}(x)\, \delta x,$$

where $x \in \mathbb{R}^{18}$ stacks $(r_1, r_2, r_3, v_1, v_2, v_3)$ and J_{f_θ} is estimated numerically (finite-difference Jacobian). We compare observed perturbation growth $\|\delta x_{\mathrm{obs}}(t)\|$ (difference of two nearby rollouts) with linearized growth $\|\delta x_{\mathrm{lin}}(t)\|$, reporting:

- finite-time Lyapunov estimates from observed and linearized growth,
- median/p90/max relative norm error between observed and linearized perturbations.

This does not prove global validity, but it quantifies whether the discovered local dynamics are self-consistent over the tested horizon.

Candidate selection now also records per-candidate redundancy flags and maximum in-model feature collinearity. Under `-redundancy-prune strict`, candidates can be dropped before ranking if they are algebraically redundant (e.g., near linear combinations of gated/base terms) or exceed a configurable collinearity threshold. With `-sensitivity-mode objective`, the ranking key is adjusted to

$$\mathrm{RMSE}_{\mathrm{rollout}} + \lambda_s \, \varepsilon_{\mathrm{sens}},$$

where $\varepsilon_{\mathrm{sens}}$ is median sensitivity relative error and λ_s is user-configurable. For exact partition identities (e.g., $g_{i,\alpha} = g_{i,\alpha}^{\mathrm{close}} + g_{i,\alpha}^{\mathrm{far}}$), we canonicalize coefficients before auditing by removing one redundant gated term:

$$c_b g + c_c g^{\mathrm{close}} + c_f g^{\mathrm{far}} \equiv (c_b + c_f) g + (c_c - c_f) g^{\mathrm{close}}.$$

3.8 Delay-embedding local-map track with sensitivity scoring (new in this revision)

To complement force-library sparse discovery, we added a deterministic scalar-channel predictor based on delay-coordinate embedding[8]. For a chosen scalar observable x_t (e.g., `a1_x`, `min_pair_dist`), we build:

$$z_t = [x_t,\ x_{t-\tau},\ \ldots,\ x_{t-(m-1)\tau}], \qquad y_t = x_{t+1}.$$

We then fit local models on k nearest neighbors in embedding space:

- **Local linear**: $y \approx \beta^\top \phi(z)$ with $\phi(z) = [1, z]$.
- **Local linear-fractional (rational)**:

$$y \approx \frac{\beta^\top \phi(z)}{1 + \delta^\top \phi(z)}.$$

Both are fitted deterministically by ridge-regularized normal equations in each neighborhood.

Selection uses strict chronological train/validation/holdout splits (with explicit no-future leakage checks for chronological mode). Primary metric is holdout MSE against the persistence baseline $\hat{y} = x_t$. We also score local sensitivity consistency: for nearby holdout pairs (z_i, y_i) and (z_j, y_j) we compare one-step observed perturbation $\Delta y_{\mathrm{obs}} = y_j - y_i$ with tangent prediction $\Delta y_{\mathrm{pred}} \approx \nabla \hat{f}(z_i) \cdot (z_j - z_i)$, and summarize relative error:

$$\varepsilon_{\mathrm{sens}} = \mathrm{median}\left(\frac{|\Delta y_{\mathrm{pred}} - \Delta y_{\mathrm{obs}}|}{\max(|\Delta y_{\mathrm{obs}}|, \epsilon)} \right).$$

Validation ranking is

$$\mathrm{rank_score} = \mathrm{MSE}_{\mathrm{val}} \left(1 + \lambda_{\mathrm{sens}} \varepsilon_{\mathrm{sens,val}}\right),$$

with user-set $\lambda_{\mathrm{sens}} \geq 0$. For `delta_mlp` candidates, report artifacts now additionally include deterministic gate diagnostics (`delta_mlp_diagnostics`): per-sensor gate weights, active-sensor fraction at a fixed threshold, effective sensor count, and the matched NN-row metrics. We also log per-model evaluation counts (`model_eval_counts`) so side-by-side model-family comparisons are auditable.

3.9 What counts as "predicts well"

Pointwise acceleration fit is not sufficient. We evaluate:

- Rollout accuracy: integrate the discovered model forward and compare trajectory error (e.g., position RMSE) against the oracle.
- Stability and failure modes (blow-ups, divergence time).
- Generalization to held-out initial conditions within the intended regime.

Editorial uncertainty: In chaotic regimes, long-horizon point prediction is not meaningful[6]; evaluation should shift toward short-horizon deterministic accuracy, ensemble/statistical predictions, and qualitative invariants and manifold structure. In the current implementation we partially address this with predictability tooling: (i) small ensembles of nearby initial conditions to measure conditioning and detect when the ensemble agrees on an emergent binary/outcome (*"lock points"*), and (ii) event-driven extraction of close-encounter minima plus a lightweight sparse baseline that models energy and angular-momentum exchange across encounters. These tools aim to quantify *when* predictions become well-conditioned, without claiming universal long-horizon microstate predictability.

3.10 LLM-assisted interpretation (optional)

We add an optional LLM judge layer to interpret candidate equations and propose next experiments. The LLM does *not* override numeric ranking; it is a supplemental reviewer that produces a structured scorecard and narrative synthesis. Concretely, it decides (a) proposed initial conditions within bounds, (b) an interpretable ranking of candidates under a fixed rubric, and (c) next-step recommendations (rollout integrator, discovery solver/hyperparameters, and search directions), while the discovery engine performs coefficient estimation/search (STLS/LASSO or an optional GA baseline). The rubric is explicit and machine-readable, with component scores for (i) predictive fidelity, (ii) parsimony, (iii) physical plausibility, (iv) regime consistency, and (v) stability risk. Evidence passed to the judge includes equation forms plus numeric metrics (MSE, rollout RMSE, divergence time) and regime diagnostics.

Optionally, the judge may also propose a concrete candidate equation written directly in the project's feature language (a vector form for $a_{i,x}$, $a_{i,y}$, $a_{i,z}$ over the currently selected feature library) and/or recommend switching between small feature libraries (`default`, `extended`, `em_fields`). These proposals are treated as hypotheses: they are parsed, validated, and scored by the same numeric evaluators before they are considered "better". We require JSON-only output to preserve determinism, and all prompts and responses are logged as reproducibility artifacts. To stabilize iterative progress, we additionally maintain a global "best so far" ledger over local runs (e.g., `results/best_results.md` and `results/best_results.json`), grouped by comparable simulation buckets (steps, dt). The factory loop can include the incumbent best equation and its metrics in the LLM context as a concrete target to beat, while keeping numeric selection as the source of truth. The judge can additionally recommend (a) which rollout integrator to use for evaluation (Euler vs. leapfrog), (b) which fitness heuristic to use for candidate selection (MSE vs. MSE with a parsimony penalty), and (c) which discovery solver/hyperparameters to try next (STLS/LASSO/GA), enabling controlled ablations without compromising reproducibility.

4 System Integration and Reproducibility

4.1 CLI workflow (main-body summary)

The implementation is organized as a reproducible CLI pipeline with four stages:

- **oracle generation** (`simulate`);
- **candidate-model generation** (`discover`, `predictability takens`);
- **strict efficacy reporting** (`predictability report`/`compare`, claim gates);
- **optional closed-loop automation** (`factory`, `quickstart`, `bench-em`).

For readability, command-level details and full artifact inventories are moved to Appendix A.

4.2 Factory loop and long-run reuse

The factory loop follows a fixed cycle: propose initial conditions, simulate, discover candidate equations, score under a fixed rubric, and continue/stop. Numeric holdout metrics remain the decision authority; optional LLM output is advisory and fully logged. To support longer searches and cross-run reuse, this revision adds recursive archive seeding: `-equation-archive-seed` now accepts archive files or directories, scans directories for prior `equation_search_archive.json` artifacts, merges node statistics, and warm-starts parent selection. This is infrastructure for longer discovery loops; measured efficacy impact is reported in Section 4 and Appendix B.

4.3 Reproducibility contract

The reproducibility contract is: fixed seeds, fixed simulator/discovery settings, structured sidecars, and explicit claim-gate artifacts. Current outputs include trajectory CSV + JSON sidecar (config, header hash, warnings, regime diagnostics, dt statistics, encounter metadata, initial state), per-iteration reports (`report.json`, `rollout_trace.json`), and end-of-run claim files (`claim_assessment.json`, `strict_holdout_report.json`, `evaluation_aggregate.json`, `sensitivity_eval.jso`

4.4 Robust I/O design

We explicitly fixed a key fragility: rigid CSV schemas break as output evolves. We therefore parse by header name and extract only needed columns for any given task. This enables rapid iteration.

4.5 Determinism and metadata

A "truth-grade" oracle requires:

- Fixed integrator and tolerance settings recorded.
- Initial condition specifications recorded.
- Constants and force toggles recorded.

Current implementation outputs CSV and a JSON "sidecar" containing config, header hash, warnings, last-regime diagnostics, dt statistics (accepted/rejected/min/max/avg), close-encounter event metadata (when triggered), and the initial state (masses/charges/positions/velocities) to support strict reproducibility.

4.6 Efficacy reporting from simulation-extracted information (new)

To align evaluation with the guiding question, we added a reporting layer that consumes per-channel predictability artifacts and classifies whether each channel shows usable information gain.

For each channel c, let $\Delta_{\mathrm{rel}}(c)$ be holdout relative improvement versus persistence and let $\varepsilon_{\mathrm{sens}}(c)$ be holdout sensitivity-median error. With thresholds τ_Δ and τ_s, we define:

$$\mathrm{effective}(c) = \mathbf{1}\{\Delta_{\mathrm{rel}}(c) > \tau_\Delta \ \wedge \ \varepsilon_{\mathrm{sens}}(c) \le \tau_s\}.$$

We then partition channels into raw-state and derived-diagnostic sets and report:

$$\Delta_{\mathrm{info}} = \mathrm{median}_{c\in\mathrm{raw}}\, \Delta_{\mathrm{rel}}(c) - \mathrm{median}_{c\in\mathrm{derived}}\, \Delta_{\mathrm{rel}}(c).$$

The CLI command `predictability report` writes machine-readable JSON plus Markdown summaries for this rubric, so claims about "better information → better math" are tied to explicit thresholds and reproducible artifacts.

5 Results (What we expect, what we can claim, and what we cannot)

5.1 Expected outcomes

For gravity-only or weak-EM regimes:

- **gravity_min** should rediscover near-inverse-square directional structure.
- **em_basic** may recover Lorentz-like cross terms if EM effects are present and sufficiently sampled.
- Sparse models can achieve strong short-horizon rollout fidelity with few terms.

5.2 Matched geometry-feature and MDL-selector ablation (February 8, 2026)

We ran two matched deterministic gravity-only factories (`research_v2_atlas`, STLS, 3 iterations, `steps`=150, $dt = 0.01$, `highbar_v2_benchmark_first`, LLM off):

- **Baseline**: feature families `newtonian,pn1,yukawa,darwin_like,tidal_invariants,jerk_augmented,hami` selector `numeric_only`.
- **New variant**: baseline + `jacobi_legendre,shape_invariants,symmetry_invariants`, selector `numeric_mdl`.

The best discovered equation text in both matched runs was unchanged (smooth-gated Newtonian decomposition with 27 terms), and no top candidate included the new Jacobi/shape/symmetry atoms.

metric	baseline	geometry+MDL
best within-run rollout RMSE	1.3160×10^{-2}	1.3160×10^{-2}
benchmark RMSE mean	$1.3561644469 \times 10^{-2}$	$1.3561644469 \times 10^{-2}$
seed-suite median relative improvement	n/a (no prior)	0.0
seed-suite bootstrap CI_{95}	n/a	[0, 0]
benchmark relative improvement	n/a	1.279×10^{-16}
strict holdout pass	false	false

Sensitivity-equation diagnostics (geometry+MDL best model): observed FTLE ≈ 6.6756, linearized FTLE ≈ 6.4739, relative perturbation-norm median ≈ 0.0648.

We then ran a targeted global-model check (`research_v1`, 1 iteration) with only `newtonian,jacobi_legendre,s` and `numeric_mdl`. That run selected a *simpler* model (complexity 5; near-Newtonian with a small smooth-gate split in z), but strict-holdout evidence was slightly negative: benchmark relative improvement -7.51×10^{-5} and seed-suite median relative improvement -4.70×10^{-5} (95% CI fully below zero).

For physicists: this is a clear negative result under a stricter novelty attempt; richer geometry features and MDL ranking did not improve holdout predictive accuracy in this bucket. For novice readers: we added smarter math features and a better "simplicity-vs-fit" chooser, but predictions did not get better on new tests.

5.3 Takens local-map + sensitivity ablation (February 8, 2026)

We ran the new delay-embedding predictor (`predictability takens`, chronological split, train/val/holdout $= 0.70/0.15/0.15$, grid $\tau \in \{1, 2, 3\}$, $m \in \{3, 4, 5\}$, $k \in \{6, 8, 12\}$, $\lambda \in \{10^{-8}, 10^{-6}, 10^{-4}\}$, `model=both`, $\lambda_{\text{sens}} = 0.1$) on four channels of `traj.csv`. This ablation intentionally excludes residual (delta) models and auxiliary sensors to isolate the behavior of linear/rational local maps.

channel	best model	holdout relative improvement vs persistence	holdout sensitivity median
`min_pair_dist`	linear	-22.3386	≈ 1.00
`max_accel`	linear	-22.3303	≈ 1.00
`a1_x`	linear	$+0.99745$	1.55×10^{-2}
`v1_x`	linear	$+0.99778$	2.06×10^{-2}

Interpretation:

- For derived extrema channels (`min_pair_dist`, `max_accel`), persistence is far stronger than the local-map candidates in this run.
- For raw state channels (`a1_x`, `v1_x`), the local deterministic map is substantially better than persistence on one-step holdout prediction.
- Rational local maps did not beat local linear maps on these trajectory channels; in additional rational-only checks, holdout MSE was slightly worse than linear for `a1_x` and `v1_x`.

For physicists: the result supports a mixed view of deterministic local formulas. Delay-embedding local models can provide strong short-horizon prediction for some observables, but they do not yet give a uniformly better reduced law across derived chaotic diagnostics. For novice readers: this new method can predict some signals much better than a naive "next value equals current value" guess, but it still fails badly on other signals, so we are not at a universal formula.

5.4 Aggregate efficacy report from extracted channels (February 8, 2026)

Using `predictability report` with deterministic bootstrap CIs on four channel reports (`a1_x`, `v1_x`, `min_pair_dist`, `max_accel`) from the delta-enabled target-only baseline (`model=all`, thresholds $\tau_\Delta = 0$, $\tau_s = 0.1$), we obtained:

- $n_{\text{channels}} = 4$, $n_{\text{effective}} = 2$ (effective rate $= 0.5$),

- median improvement (all channels) ≈ 0.23797 with $\mathrm{CI}_{95} \approx [-0.52325, 0.99981]$,
- median improvement (raw) ≈ 0.99950 with $\mathrm{CI}_{95} \approx [0.99919, 0.99981]$,
- median improvement (derived) ≈ -0.52325 with $\mathrm{CI}_{95} \approx [-0.52325, -0.52325]$,
- Δ_{info} (raw minus derived) ≈ 1.52275,
- claim status: `information_helpful_in_some_channels`.

Interpretation: residual local formulas alone are not sufficient for uniform improvement; without auxiliary sensors, derived diagnostics remained below persistence in this matched run.

5.5 Sensor-augmented efficacy comparison (February 8, 2026)

We then ran a matched baseline-vs-sensor comparison on a fresh trajectory (`steps`=800, $dt = 0.01$), keeping the same strict holdout protocol and delta-enabled model search (`model=all`):

- **Baseline**: Takens uses only the target channel itself.
- **Sensor-augmented**: Takens uses additional simulator channels via `-sensors` (state components for `a1_x`, `v1_x`, `min_pair_dist`; acceleration components for `max_accel`).

Aggregate efficacy outcomes over the same four targets (`a1_x`, `v1_x`, `min_pair_dist`, `max_accel`):

- baseline: median all ≈ 0.23797, effective channels $= 2/4$;
- sensor-augmented: median all ≈ 0.99989 with $\mathrm{CI}_{95} \approx [0.99980, 0.99999]$, median raw ≈ 0.99998, median derived ≈ 0.99980, effective channels $= 4/4$.

Using `predictability compare` (`before=efficacy_delta_baseline`, `after=efficacy_delta_sensor_augmen`
we measured:

- all-channel median improvement delta $\approx +0.76192$,
- derived-channel median improvement delta $\approx +1.52305$,
- effective-channel delta $= +2$ (from 2 to 4),
- channel non-regression rate $= 1.0$ and `overall_non_regression=true`.

Interpretation for the guiding question: in this controlled matched run, richer simulator information *plus* an appropriate deterministic equation class (residual local maps) did yield materially better strict-holdout prediction across all tested channels. For physicists: this is strong one-step evidence for representation matching (state augmentation + residualization), not a claim of long-horizon shadowing. For novice readers: once we gave the equation better inputs and asked it to predict the change instead of the full value, it became much more accurate on every test signal we checked here.

5.6 Predictability context-window ladder (February 8, 2026)

We implemented `predictability context-window` to sweep embedding context length and estimate minimum effective context under strict holdout gates. For each point, context volume is $W = (\tau m)|S|$.

Matched runs used `tau=1`, $m \in \{1, 2, 3, 4, 5, 6, 8, 10\}$, `k` in $\{6, 10\}$, λ in $\{10^{-8}, 10^{-6}\}$, `model=all`, chronological split, and a high-bar gate $\Delta_{\text{rel}} > 0.6$, sensitivity median ≤ 0.2.

regime	target/sensors	estimated $W^\star$	note
one-body inertial	`r1_x` / `r1_x,v1_x,a1_x`	3	effective already at $m = 1$
two-body Kepler	`r1_x` / `r1_x,v1_x,a1_x`	3	effective already at $m = 1$
three-body perturbed	`a1_x` / `a1_x,v1_x,r1_x,r2_x,r3_x`	20	first high-bar pass at $m = 4$

Artifacts: `results/context_window_one_body.json`, `results/context_window_two_body.json`, `results/context_window_three_body_a1x.json`, and ladder summary `results/context_window_ladder.json`.

Interpretation:

- Under this high-bar gate, the measured effective context requirement is substantially larger for the perturbed three-body regime than for one-body/two-body regimes.
- This supports a practical "information-context law" view: deterministic dynamics are not random, but the information/context budget needed for reliable local prediction can increase sharply with interaction complexity.
- Negative result (important): for perturbed three-body `r1_x` with a smaller sensor set (`r1_x,v1_x,a1_x`), no effective point was found in the tested grid even under a weaker gate ($\Delta_{\text{rel}} > 0.2$, sensitivity ≤ 0.2), so $W^\star$ was "not found" in that setting.
- We also observed non-monotonic performance across m in the three-body case; so increasing context alone is not guaranteed to improve accuracy without matching model class and feature geometry.

5.7 Follow-up sensor-context stress test (February 8, 2026)

To chip away at the "not yet claimed" items with additional strict evidence, we ran a same-day follow-up trajectory experiment (`three-body-perturbed`, truth mode, `steps`=800, $dt = 0.01$) and evaluated two targeted scans under the same high-bar gate ($\Delta_{\text{rel}} > 0.6$, sensitivity median ≤ 0.2).

A1) Fixed-context sensor-count check on `a1_x`. With fixed context (`tau`=1, `m`=2) and `model=all`, all tested sensor sets failed:

sensor set	$\lvert S\rvert$	context volume W	holdout relative improvement
`local5` (`a1_x,v1_x,r1_x,r1_y,r1_z`)	5	10	−61.52
`coupled5` (`a1_x,v1_x,r1_x,r2_x,r3_x`)	5	10	−67.81
`rich19` (target + full r/v state)	19	38	−76.69

Sensitivity medians remained below 0.2 in all three cases, so failure was driven by poor holdout accuracy rather than tangent-consistency instability.

A2) Channel-specific context scan on `min_pair_dist`. Using `delta_rational` and sweeping `m=1..4`:

sensor set	best Δ_{rel} in tested grid	any high-bar pass?	estimated $W^\star$
`target1` (`min_pair_dist` only)	$+0.7183$ (at $m = 4$)	yes	4
`pos10` (target + positions)	-2.4827	no	not found
`state19` (target + full r/v state)	$+0.4367$	no	not found

Interpretation for the guiding question:

- This follow-up produced one new strict positive point (`min_pair_dist`, target-only, $W^\star = 4$ in tested grid), so we did chip away at the "no strict pass in this channel family" concern.
- More sensors were not automatically better in this setup; richer sensor sets could reduce or erase strict holdout gains for the same target/model family.
- Therefore, "better information" should be interpreted as *better-matched information*, not simply larger sensor count.

Artifacts: `results/exp_sensor_context_20260208_133940/sensor_information_value_m2_summary.json` and `results/exp_sensor_context_20260208_133940/min_pair_dist_sensor_context_scan_summary.json`.

5.8 Temporal neural baseline update (February 10, 2026)

We upgraded the neural path by separating two deterministic residual baselines:

- `delta_mlp`: gated delay-window residual MLP.
- `delta_tcn`: distinct temporal-convolutional residual baseline (no longer an alias).

Both models use sparse sensor gates and residual prediction,

$$\hat{x}_{t+1} = x_t + \widehat{\Delta x}_{t+1}.$$

We also added multi-horizon recursive holdout metrics ($h \in \{1, 2, 4, 8\}$) to Takens reports (when horizon recursion is well-defined for the selected sensor setup).

Matched head-to-head runs on `traj.csv` (`a1_x`, same search grid and split protocol, model fixed per run):

- **Single-sensor** (`sensors=a1_x`): `delta_mlp` achieved relative improvement ≈ 0.98856 (sensitivity median ≈ 0.03563), while `delta_tcn` achieved ≈ 0.99085 (sensitivity median ≈ 0.02699). Paired efficacy comparison gave $\Delta_{\text{rel}} \approx +0.00229$ in favor of `delta_tcn` with non-regression true.
- **Sensor-augmented** (`sensors=a1_x,v1_x,v2_x,min_pair_dist`): `delta_mlp` achieved relative improvement ≈ 0.97633 (sensitivity median ≈ 0.05771), while `delta_tcn` achieved ≈ 0.99645 (sensitivity median ≈ 0.02921). Paired efficacy comparison gave $\Delta_{\text{rel}} \approx +0.02012$ in favor of `delta_tcn` with non-regression true.

For the single-sensor `delta_tcn` run, recursive horizon metrics remained strongly positive: relative improvement ≈ 0.9909 (1-step), 0.9902 (2-step), 0.9895 (4-step), and 0.9938 (8-step). This is still a finite-horizon result on one channel, not a global chaos-resolution claim.

Reporting now explicitly separates evidence tiers. In these runs, efficacy aggregate output was `evidence_tier=predictive_signal` and `science_claim_allowed=false`: predictive gain was strong on the tested channel, but insufficient in breadth for a physics-grade novelty claim. Artifacts: `results/novelty_20260210_headtohead/takens_a1x_single_delta_mlp.json`, `results/novelty_20260210_he` `results/novelty_20260210_headtohead/takens_a1x_aug_delta_mlp.json`, `results/novelty_20260210_headt` and paired comparisons in `compare_single_mlp_vs_tcn.json` and `compare_aug_mlp_vs_tcn.json`.

5.9 Policy and claim-gate summary (main-body view)

Extended policy sweeps and paired matrix studies are now treated as supporting evidence rather than core narrative and are reported in Appendix B. In the main body, the key outcomes are:

- Archive/MCTS policy tuning improved search bookkeeping and diversity but did not produce robust cross-seed predictive gains.
- In global-equation policy sweeps, the strict claim gate remained conservative, emitting `claim_status=no_impr` whenever holdout evidence was not statistically positive.
- In both gravity-dominant and EM-active paired matrices, policy updates were mostly neutral and occasionally worse on held-out benchmarks.

This behavior is aligned with the paper's guiding question: we only claim progress when richer information plus new math improves strict holdout prediction.

5.10 Cross-run archive seeding for longer loops (February 10, 2026)

To enable longer search loops that reuse prior candidates, we added recursive archive seeding in `factory` via `-equation-archive-seed`. This allows a new run to import prior equation archives and start selection from accumulated historical candidates.

We evaluated a matched-seed gravity-only pair (`research_v2_atlas`, 4 iterations, `steps`=180, $dt = 0.01$, STLS, mock LLM, strict claim gate):

metric	baseline	seeded (matched seed)
iteration-1 top rollout RMSE	$5.713546713 \times 10^{-2}$	$5.713546707 \times 10^{-2}$
final-iteration top rollout RMSE	$1.501334298 \times 10^{-2}$	$1.501438012 \times 10^{-2}$
benchmark RMSE mean (3-case suite)	$1.698633932 \times 10^{-2}$	$1.698756042 \times 10^{-2}$
claim status	baseline-only (`n_cases`=0)	`no_improvement` (`n_cases`=12)
strict holdout pass	false	false

In this matched comparison, archive seeding was essentially neutral at iteration 1 and slightly worse at final/benchmark metrics (relative changes $\approx 10^{-9}$ at iteration 1 and $\approx 6.9 \times 10^{-5}$ at final RMSE). In an additional different-seed seeded run, iteration-1 error improved, but final strict-holdout outcomes remained non-improving. Therefore, current evidence supports archive seeding as a useful search-memory mechanism, not yet as a demonstrated strict-holdout accuracy improvement.

5.11 What we can claim

- The simulator provides consistent numerical solutions to the implemented equations with user-controlled accuracy (RK45 mode).

- The discovery engine can identify compact, interpretable approximations within structured libraries and validate them by rollout.
- The archive-guided search now reliably preserves, ranks, and reuses equation structures across iterations and across runs (via recursive archive seeding), even when strict-holdout accuracy gains are not yet demonstrated.
- The reporting pipeline now separates evidence tiers (`engineering_only`, `predictive_signal`, `physics_candidate`) and exposes an explicit `science_claim_allowed` flag.
- The claim-gated reporting pipeline remains conservative in practice: when improvements are absent or unproven, it emits `no_improvement` rather than optimistic language.

5.12 What we cannot honestly claim (yet)

- That the simulator is "physically exact" for EM in all regimes (it is quasi-static).
- That energy conservation is strict in combined EM dynamics as implemented (field energy not modeled).
- That close-encounter accuracy is truth-grade without regularization/high precision.
- That a single global simple formula predicts chaotic three-body dynamics long-horizon.

6 Limitations and Open Questions (Editorial)

This section is intentionally frank.

Electromagnetic modeling fidelity

The quasi-static magnetic approximation may be adequate for some regimes but fails when retardation or radiation are significant. If the objective is "physics truth," the EM model must be upgraded (retarded potentials or field solver), at significant computational cost.

Integrator structure under EM

Leapfrog is attractive for gravity; under EM it may not preserve the right structure. We currently treat leapfrog as a fast option, not as a guaranteed long-term invariant-preserving method under full coupled dynamics. EM-appropriate schemes include the Boris pusher and structure-preserving charged-particle integrators.[17, 18, 19]

Close encounters and stiffness

Softening (even when smoothed) is not a principled solution. We also support deterministic substepping near close encounters, but a truly truth-grade approach likely requires regularization (Sundman transform, KS/chain regularization) and possibly higher precision.

Discovery stability and identifiability

Sparse regression can select misleading terms when features are correlated or data is not sufficiently rich. Better experimental design (diverse initial conditions, active learning) and more robust optimization (e.g., constrained solvers) are likely necessary. In practice, a common failure mode is to repeatedly rediscover gravity-only structure even when EM is enabled. We therefore log a simple identifiability diagnostic on each oracle trajectory: the ratio $\mathbb{E}(|a_{\mathrm{em}}|)/\mathbb{E}(|a_{\mathrm{grav}}|)$. If this ratio is $\ll 1$, EM terms are unlikely to be selected by sparse regression under the chosen regimes and noise levels. To operationalize "rich enough" data, we include a deterministic EM benchmark suite (`bench-em`) that sweeps charge scale and velocity scale across multiple initial-condition templates to produce EM-heavy regimes where EM terms should become identifiable.

Meaning of "novel math"

We interpret "novel" as compact approximations that outperform naive baselines within a regime and are expressed in human-interpretable terms. We do not claim a universal analytic solution.

7 Future Work

Future work priorities are explicitly ordered by the guiding question posed by Jonathan Hendler: whether richer information plus new mathematics yields better strict-holdout prediction over time.

- Promote residual local maps and sensor-augmented embeddings to the default predictability benchmark path (done for model-family default; remaining work: broader channel suites and multi-seed replication).
- Turn the context-window hypothesis into a full response surface $W^\star(\text{regime}, \text{channel}, \tau_\Delta, \tau_s, H)$, including multi-horizon H and uncertainty bands over seeds.
- Expand strict multi-horizon holdout metrics (1-step, short-horizon, and shadowing-time) from single-channel diagnostics to broader channel/regime suites so "predicts better over time" is measured directly, not inferred from one-step scores.
- Build explicit information-value experiments: compare sensor subsets under matched splits and compute uncertainty bars for each subset to estimate marginal value of each diagnostic channel.
- Strengthen the new neural path with physics-grade constraints: longer delay windows (TCN/attention variants), symplectic or Hamiltonian-inspired regularization, and calibrated sparse-gate learning validated by strict holdout ablations.
- Add deterministic sensor-selection and shrinkage controls (e.g., sparse sensor masks and complexity penalties on embedding channels) so "more sensors" does not automatically mean worse conditioning or overfit.
- Extend deterministic equation families with mathematically compact non-physics bases from dynamical systems: Koopman/EDMD dictionaries[11], Ulam transfer operators, symbolic-dynamics/entropy features, and recurrence-quantification gates.
- Keep strict falsification gates: paired before/after comparisons with non-regression flags, bootstrap CIs, and preregistered thresholds before any novelty claim.

- Improve oracle fidelity where it most limits discovery: close-encounter regularization (KS/chain or Sundman), higher-precision truth mode, and stronger EM-model caveat tracking.
- Auto-generate paper tables/figures directly from JSON artifacts to prevent transcription drift between code outputs and manuscript claims.

8 Conclusion

Direct answer to the primary question

Yes, in a qualified sense. Under strict one-step holdout evaluation, richer simulator information plus a better local representation (sensor-augmented residual maps) improved matched efficacy from 2/4 effective channels to 4/4, with paired non-regression on all shared channels. Follow-up stress tests also showed the converse: adding sensors can hurt when channel/model matching is poor. Therefore, better information can enable better predictive mathematics in a deterministic three-body setting, but gains are channel- and representation-dependent.

Formal claims supported by current evidence

- **Claim C1 (local predictability gain):** In the tested matched protocol, sensor-augmented delta-enabled Takens models materially improved holdout performance versus baseline local context.
- **Claim C2 (context-demand signal):** Defining context volume $W = (\tau m)|S|$, the estimated minimum effective context under a high-bar gate increased from $W^\star = 3$ (one-body/two-body references) to $W^\star = 20$ (perturbed three-body, tested channel/sensors). This supports an *empirical* information-context scaling hypothesis.
- **Claim C3 (negative global result):** In global force-library discovery, added geometry families and MDL-aware ranking did not yield robust strict-holdout gains over the Newtonian baseline in reported matched runs.
- **Claim C4 (non-monotonic sensor value):** In follow-up strict scans on `min_pair_dist`, a target-only embedding reached a high-bar pass at $W^\star = 4$, while richer position/state sensor sets did not pass in the tested grid. Information value is therefore not monotonic in sensor count.
- **Claim C5 (current temporal NN directional result):** In matched `delta_mlp` vs. `delta_tcn` Takens runs on `a1_x`, `delta_tcn` improved strict-holdout relative improvement in both tested sensor settings (single-sensor and sensor-augmented), with paired non-regression true.
- **Claim C6 (search-memory result):** Recursive archive seeding enables cross-run candidate reuse and warm-start parent selection for longer loops, but in the current matched test it did not improve strict holdout metrics.

What is genuinely novel in this work

- Not a new universal law, but a reproducible *novelty gate*: claims are conditioned on strict holdout improvement, sensitivity consistency, and paired non-regression.

- An explicit information-context metric and testable threshold workflow ($W^\star$) integrated into routine reporting, building on embedding theory but operationalized for this simulator/discovery stack[8, 9, 10].
- A deterministic sensor-value protocol that quantifies when richer measurements improve interpretable local formulas, complementing prior three-body ML-surrogate work[7].

What is not concluded

- We do *not* claim a universal closed-form three-body law.
- We do *not* claim long-horizon shadowing success from one-step metrics.
- We do *not* claim a proved theorem for $W^\star$ scaling; current evidence is empirical, protocol-dependent, and channel-dependent.
- We do *not* claim that adding more sensors is always beneficial; current evidence shows channel-dependent gains and failures.
- We do *not* claim that neural baselines are uniformly superior across channels or regimes; current neural gains are channel-specific and finite-horizon.

Claim gate for future novelty statements

To keep conclusions scientifically tight, future "novel math" claims should satisfy all of:

$$\Delta_{\mathrm{rel}} > 0, \quad \varepsilon_{\mathrm{sens}} \leq \tau_s, \quad \text{paired non-regression} = \texttt{true},$$

plus uncertainty support (bootstrap CI not crossing zero in the relevant direction) on preregistered holdout suites.

Practical takeaway

For physicists: current novelty is regime-specific and finite-horizon; local residual maps with explicit sensor context are promising, while global compact closures remain unresolved. For novice readers: the system is chaotic but not random; better measurements and smarter short-term equations can improve predictions, yet one simple formula for all long-term behavior remains out of reach.

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A Implementation Notes (brief)

- Main body now keeps only workflow-level reproducibility summary; command-by-command operational detail is intentionally kept in appendices.
- CSV parsing must be header-based to remain stable as columns evolve.
- Discovery expands vector bases into x/y/z component features and estimates sparse linear combinations via STLS/LASSO sparse regression (with an optional GA baseline).
- Factory now includes a charged preset (`em-three-body`) so EM-enabled policy comparisons can be run without external IC injection.
- Factory now persists equation-search state (`equation_search_archive.json`), uncertainty-aware ranking metadata, and a compact archive report (`equation_search_report.md`) for auditability.
- Factory supports recursive cross-run archive seeding via `-equation-archive-seed`: archive files or directories are scanned for `equation_search_archive.json`, merged into current archive state, and logged in `equation_search_archive_seeded_from.json`.
- Factory policy/gating metadata is explicit (`policy_effective.json`); optional publication artifacts include `evaluation_aggregate.json`, `claim_assessment.json`, `strict_holdout_report.json`, `sensitivity_eval.json`, and `paper_summary.md`.
- Advanced controls now include multi-smooth atlas variants, one-shot smooth gate-geometry tuning (`gate_geometry.json`), strict redundancy/collinearity pruning with algebraic partition canonicalization, sensitivity-aware ranking, selectable feature-family hypotheses, and deterministic equation-template injection for non-Newtonian candidates.
- New deterministic feature-family atoms include `jacobi_legendre`, `shape_invariants`, and `symmetry_invariants`; selection can be switched to an MDL-prioritized comparator via `-selector-policy numeric_mdl`.

- The `predictability takens` command now supports local linear, local rational, and residual (delta) model families (`-model`, default `all`), sensor-augmented embeddings from simulator columns (`-sensors`), strict split modes, and sensitivity-weighted selection (`-sensitivity-weight`), producing JSON reports with holdout and tangent-consistency metrics.
- Neural Takens baselines now include distinct `delta_mlp` and `delta_tcn` model kinds; `delta_tcn` is no longer an alias to `delta_mlp`.
- Takens reports now include recursive multi-horizon holdout metrics (`best.horizon_metrics`) when the selected setup supports deterministic recursion.
- Takens best-model outputs now include explicit architecture metadata (`model_family`, `depth`, `width`, `feature_dim`) and parameter counts (`parameter_count`) for reproducible model-complexity reporting.
- The `predictability report` command aggregates channel-level efficacy from report JSON files with explicit thresholds (`-improvement-threshold`, `-max-sensitivity-median`), deterministic bootstrap confidence intervals (`-bootstrap-*`), and JSON/Markdown outputs. The `predictability compare` command provides paired before/after effect-size and non-regression summaries.
- Efficacy aggregates now expose explicit evidence-tier fields (`evidence_tier`, `science_claim_allowed`) to separate engineering progress from claim-grade scientific evidence.
- The `predictability context-window` command sweeps embedding context length (`m`) at fixed delay (`tau`) and reports minimum effective context under strict holdout+sensitivity gates, including per-point artifacts and JSON/Markdown summaries.
- New deterministic presets include `one-body` and `three-body-perturbed` for complexity-ladder experiments.
- The `research_v2_atlas` policy defaults to an atlas model family and benchmark-first claim gating (`highbar_v2_benchmark_first`).
- Bootstrap CI computation for the claim gate now uses index-safe resampling and is covered by a regression test to prevent out-of-bounds failures on 12-case suites.

B Extended Policy and Matrix Evidence

B.1 Archive-guided search sweep details

We ran a controlled local sweep (8 factory iterations, STLS, `three-body` preset, `steps`=200, $dt = 0.01$, mock LLM), varying $c \in \{0.4, 0.7, 1.0\}$ and $K \in \{6, 8, 12\}$ at fixed seed.

- Across all settings, the best rollout RMSE found in the run was identical: 9.2602898×10^{-3}.
- The tuned setting ($c = 0.4$, $K = 12$) was slightly better on that seed in average per-iteration top-candidate RMSE (2.7002408×10^{-2} vs. 2.7074579×10^{-2} for prior defaults $c = 0.7$, $K = 8$), and in final-iteration top RMSE (1.7625853×10^{-2} vs. 1.7921833×10^{-2}).
- The archive became more diverse (node count 38 $\rightarrow$ 45, updates 60 $\rightarrow$ 69 for the same 8-iteration budget).

Cross-seed verification (two additional seeds) showed no measurable predictive advantage between prior defaults and tuned settings; best and mean rollout RMSE were numerically identical in those checks.

B.2 Cross-run archive-seeding details (long-loop infrastructure)

On February 10, 2026, we ran a controlled archive-reuse experiment under `research_v2_atlas` (gravity-only, STLS, `steps`=180, $dt = 0.01$, 4 factory iterations, mock LLM). Three runs were compared:

- **baseline**: seed 4401, no archive seeding.
- **seeded**: seed 4417, archive-seeded from the experiment root (different-seed exploratory run).
- **seeded_match**: seed 4401, archive-seeded from the same experiment root (matched-seed fairness run).

run	iter-1 top RMSE	final top RMSE	benchmark mean RMSE	claim st
baseline	$5.713546713 \times 10^{-2}$	$1.501334298 \times 10^{-2}$	$1.698633932 \times 10^{-2}$	baseline-
seeded (seed 4417)	$4.698068919 \times 10^{-2}$	$2.071624485 \times 10^{-2}$	$1.698625189 \times 10^{-2}$	no_improve
seeded_match (seed 4401)	$5.713546707 \times 10^{-2}$	$1.501438012 \times 10^{-2}$	$1.698756042 \times 10^{-2}$	no_improve

Interpretation:

- Different-seed seeding can improve early-iteration quality, but this is not sufficient evidence of causal gain.
- In the matched-seed comparison, iteration-1 performance was effectively unchanged and final performance was slightly worse; strict holdout remained false.
- Archive seeding therefore currently functions as search-memory infrastructure (candidate carryover and warm-starting), not as a validated strict-holdout accuracy improvement.

Seed-merge metadata confirms recursive ingestion: in `seeded_match`, 40 nodes were added from baseline and 36 additional nodes were added from the exploratory seeded run (with 40 merged existing nodes), resulting in 76 post-seed archive nodes.

B.3 Claim-gate status snapshot

In the latest matched geometry+MDL run on February 8, 2026 (`research_v2_atlas`), `highbar_v2_benchmark_firs` returned:

- `claim_status=no_improvement`,
- `n_cases=12` (seed suite fully evaluated),
- `benchmark_passed=true` with essentially zero delta,
- `strict_holdout_passed=false`.

No statistically positive holdout gain implies no novelty claim, even when benchmark non-regression passes.

B.4 Claim-gated legacy vs. research matrix (gravity-dominant)

On February 8, 2026, we ran a 5-case paired matrix with clean buckets (3 factory iterations per run, STLS, LLM off, deterministic 12-seed claim suite). For each case, we first ran `legacy`, then `research_v1` with the same seed and bucket.

case	seed	steps	$\text{RMSE}_{\text{legacy}}$	$\text{RMSE}_{\text{research}}$	Δ (research-prior)
1	5151	180	1.0141×10^{-2}	1.0141×10^{-2}	0.0
2	6161	181	1.2134×10^{-2}	1.2134×10^{-2}	0.0
3	7171	182	2.0797×10^{-2}	2.0797×10^{-2}	0.0
4	8181	183	1.1167×10^{-2}	1.1355×10^{-2}	1.8847×10^{-4}
5	9191	184	1.1744×10^{-2}	1.1744×10^{-2}	≈ 0

Aggregate outcomes:

- Mean Δ (research-prior) $= 3.77 \times 10^{-5}$, median $\Delta = 0$.
- 4/5 cases were exactly equal within floating-point tolerance; 1/5 was slightly worse.
- Claim gate output was `no_improvement` in 5/5 cases, with 0/5 high-bar passes.
- In all 5 cases, the top equation text selected by `research_v1` matched the prior baseline equation.

B.5 EM-active paired matrix

To test a genuinely EM-active setting, on February 8, 2026 we ran a second 5-case paired matrix with `-em`, a charged preset (`em-three-body`), and the EM-oriented feature library (`em_fields`); each pair again used identical seed and bucket, with `legacy` first and `research_v1` second.

The mean EM-to-gravity acceleration ratio on the research best runs was 1.605 (range $[1.603, 1.607]$), so these trajectories were not gravity-dominated.

case	seed	Δ RMSE (research-prior)	Δ benchmark RMSE	claim status
1	2301	0.0	0.0	no_improvement
2	2401	-1.528×10^{-2}	$+5.001 \times 10^{-1}$	directional_trend
3	2501	≈ 0	0.0	no_improvement
4	2601	-8.013×10^{-4}	0.0	no_improvement
5	2701	0.0	0.0	no_improvement

Aggregate outcomes:

- Mean Δ RMSE (research-prior) $= -3.22 \times 10^{-3}$, median $\Delta \approx 0$.
- Mean Δ benchmark RMSE $= +1.00 \times 10^{-1}$, median $\Delta = 0$.
- Claim gate output: 4/5 `no_improvement`, 1/5 `directional_trend`, and 0/5 high-bar passes.

C Main-Body Compression Status

This revision continues the structural change requested by the project lead:

- Main body now focuses on the primary research question, methods, and claim-level results.
- Detailed build-up, expanded matrix evidence, and command/artifact implementation specifics are now concentrated in appendices.
- Remaining compression work is to move additional derivation-level and engineering-level detail out of Sections 2–3 while preserving reproducibility.